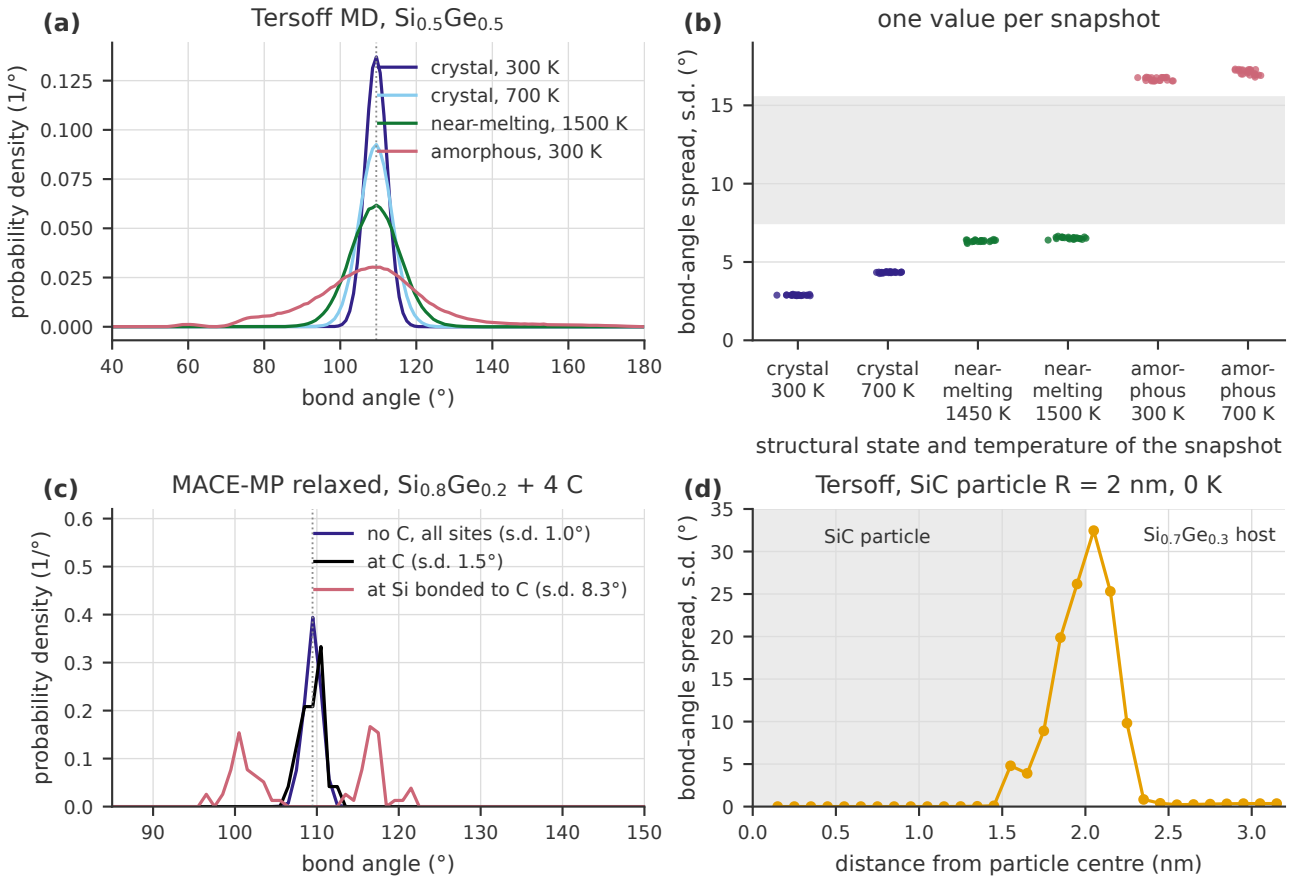
(a)
Tersoff MD, Si0.5Ge0.5
crystal, 300 K
crystal, 700 K
near-melting, 1500 K
amorphous, 300 K
probability density (1/°)
0.125
0.100
0.075
0.050
0.025
0.000
40
60
80
100
120
140
160
180
bond angle (°)
(b)
one value per snapshot
bond-angle spread, s.d. (°)
15
10
5
0
crystal 300 K
crystal 700 K
near-melting 1450 K
near-melting 1500 K
amor-phous 300 K
amor-phous 700 K
structural state and temperature of the snapshot
(c)
MACE-MP relaxed, Si0.8Ge0.2 + 4 C
no C, all sites (s.d. 1.0°)
at C (s.d. 1.5°)
at Si bonded to C (s.d. 8.3°)
probability density (1/°)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
90
100
110
120
130
140
150
bond angle (°)
(d)
Tersoff, SiC particle R = 2 nm, 0 K
SiC particle
Si0.7Ge0.3 host
bond-angle spread, s.d. (°)
35
30
25
20
15
10
5
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
distance from particle centre (nm)